CSI 4124 / SYS 5110 Foundation of Modelling and Simulation Fall 2026

Assignment #2

Instructor: Guillaume St-Onge **Due 4 November 2026, before class**

Teams of 3–4 students Worth 7% (CSI 4124) • 5% (SYS 5110) Total: 60 points

Stochastic models and their deterministic limit

In the first portion of the course, we used deterministic models to describe population dynamics, tracking the *average* number of individuals, prey or predators for instance. Real populations, made of discrete entities, are better described by stochastic models. The state of the system is then a random variable $\hat{\boldsymbol{x}}(t)$, whose realizations $\boldsymbol{x}$ are vectors of integers that change by discrete jumps when *events* (an infection, a birth, a death, …) happen at random times. In this assignment, the goal is simulating such stochastic models with Gillespie's method, and compare the results with their deterministic counterpart.

For each model below is provided the description of the state variable(s) and the potential events. Recall Gillespie's (direct) method seen in class: from the realization $\boldsymbol{x}$ at time t, compute the rate $w_k(\boldsymbol{x})$ of each event k and the total rate $W(\boldsymbol{x}) = \sum_k w_k(\boldsymbol{x})$; the time to the next event is an exponential random variable with mean $1/W(\boldsymbol{x})$, and the next event is event k with probability $w_k(\boldsymbol{x})/W(\boldsymbol{x})$.

Advice. Gillespie simulations can take a few minutes for large populations and time horizons. Start with modest settings (e.g., 5 trajectories instead of 50) to test your code, and work your way up once you are confident it is correct.

Part 1 — The stochastic SIS model

We revisit the Susceptible–Infectious–Susceptible (SIS) model of Assignment 1: in a population of fixed size N, each individual is either susceptible or infectious, and recovery confers no immunity. When there are x infectious and $N - x$ susceptible individuals, there are $x(N - x)$ susceptible–infectious pairs; each infectious individual transmits the disease to each susceptible individual independently at rate β/N, and each infectious individual recovers at rate α.

State. Let $\hat{x}(t)$ be the random variable for the number of infectious individuals at time t. Its realizations are $x \in \{0, 1, \ldots, N\}$, and the number of susceptible individuals is $N - x$.

Events.

- Event 1: infection of a susceptible individual.
  - Rate: $w_1(x) = \beta x(N - x)/N$.
  - State variation: $\Delta x = +1$.
- Event 2: recovery of an infectious individual.
  - Rate: $w_2(x) = \alpha x$.
  - State variation: $\Delta x = -1$.

Note that $x = 0$ is an *absorbing state*: once there are no infectious individuals left, no other event will happen.

Deterministic limit. In the limit of a large population, the average number of infectious individuals $i(t)$ follows the ODE of Assignment 1,

$$\frac{\mathrm{d}i}{\mathrm{d}t} = \beta \frac{i\,(N-i)}{N} - \alpha i, \tag{1}$$

where the average number of susceptible individuals was replaced with $N - i$. You can integrate Eq. (1) with your RK4 integrator from Assignment 1 or your favorite SciPy integrator. Unless a question says otherwise, use

$$\alpha = 1, \qquad \beta = 1.5, \qquad x(0) = i(0) = N/10, \qquad T = 20, \tag{2}$$

where T is the time horizon of the simulations.

A2.1 Stochastic simulation of the SIS model (30 points)

a. Implement Gillespie's method for the stochastic SIS model: create a function that returns a *single* trajectory, taking as arguments the initial condition $x(0)$, the horizon T, a random number generator and the model parameters. The trajectory consists of the times of the events $\{t_k\}$ together with the state right after each event $\{x_k\}$, starting with $(t_0, x_0) = (0, x(0))$.

We suggest the following structure, which will make your code easy to reuse in Part 2:

i. define a (population-level) rate function for each event, taking as arguments the state x and the model parameters;

ii. define the state variation Δx of each event;

iii. put these rate functions and state variations in lists or dictionaries;

iv. pass these lists or dictionaries as arguments of your function, so that the same function can simulate any model (you can then reuse it in Part 2);

v. use `rng.choice` to sample the next event from a discrete distribution with specific probabilities (here, each event is chosen with probability proportional to its rate).

(10 points)

b. For $N = 500$, generate 50 trajectories and draw them all on the same figure (a "spaghetti plot") using `plt.step` with `where="post"`, and with transparency (`alpha`) so that the individual trajectories remain visible. Overlay the deterministic trajectory $i(t)$ obtained by integrating Eq. (1). Briefly comment on the stochastic trajectories compared with the deterministic one. (7 points)

c. From your 50 trajectories, extract a sample of $\hat{x}(t^*)$, the number of infectious individuals at time $t^* = 15$. Plot its histogram and evaluate its sample mean and sample standard deviation. Identify the deterministic value $i(t^*)$ on the histogram with `ax.axvline`. Briefly comment on the sample mean compared with the deterministic value $i(t^*)$. (6 points)

Hint. For one trajectory, the state at time t^* is the state right after the last event before t^*.

d. Repeat parts b and c for $N = 5000$, producing the same figures (spaghetti plot and histogram). Then, for both values of N, plot on the same figure the histograms of the rescaled variable $\hat{x}(t^*)/i(t^*)$, and calculate its sample standard deviation. What can you say about the agreement between the deterministic and stochastic models as N grows? (7 points)

Part 2 — Stochastic predator–prey dynamics

Consider now the number of prey and predators in a closed system. Prey are born and die naturally, predators die naturally, and there are two types of predation events: in the first one, a predator eats a prey and no predator is born; in the second one, a predator eats a prey and a new predator is born.

State. Let $\hat{x}(t)$ and $\hat{y}(t)$ be the random variables for the number of prey and predators at time t, respectively. The realizations of the state $(\hat{x}, \hat{y})$ are pairs of non-negative integers, $(x, y) \in \mathbb{N}_0^2$, where $\mathbb{N}_0 = \{0, 1, 2, \dots\}$.

Events.

- Event 1: natural birth of a prey.
  - Rate: $w_1(x, y) = \alpha x$.
  - State variation: $\Delta x = +1$, $\Delta y = 0$.
- Event 2: natural death of a prey.
  - Rate: $w_2(x, y) = \sigma x$.
  - State variation: $\Delta x = -1$, $\Delta y = 0$.
- Event 3: a predator eats a prey, no predator is born.
  - Rate: $w_3(x, y) = \lambda xy$.
  - State variation: $\Delta x = -1$, $\Delta y = 0$.
- Event 4: a predator eats a prey, a new predator is born.
  - Rate: $w_4(x, y) = \delta xy$.
  - State variation: $\Delta x = -1$, $\Delta y = +1$.
- Event 5: natural death of a predator.
  - Rate: $w_5(x, y) = \gamma y$.
  - State variation: $\Delta x = 0$, $\Delta y = -1$.

Deterministic limit. In the limit of large populations, the average numbers of prey $x(t)$ and predators $y(t)$ follow the Lotka–Volterra equations

$$\frac{\mathrm{d}x}{\mathrm{d}t} = (\alpha - \sigma)\, x - (\lambda + \delta)\, xy, \qquad \frac{\mathrm{d}y}{\mathrm{d}t} = \delta\, xy - \gamma y, \tag{3}$$

which you can integrate with your RK4 integrator from Assignment 1 or your favorite SciPy integrator. Unless a question says otherwise, use

$$\alpha = 4, \quad \sigma = 1, \quad \lambda = \delta = 10^{-3}, \quad \gamma = 1, \quad x(0) = 500, \quad y(0) = 1000, \quad T = 16. \tag{4}$$

A2.2 Stochastic simulation of predator–prey dynamics (30 points)

a. Adapt your implementation of Gillespie's method to the stochastic predator–prey model. If you followed the structure suggested in Problem A2.1, only the rate functions and state variations need to change. (10 points)

b. Generate 50 trajectories and draw the spaghetti plots for the prey and for the predators (one panel each), as in Problem A2.1. Overlay the deterministic trajectories $x(t)$ and $y(t)$ obtained by integrating Eq. (3). Briefly comment on the stochastic trajectories compared with the deterministic ones. (7 points)

c. Extract samples of $\hat{x}(t^*)$ and $\hat{y}(t^*)$ at $t^* = 3.79$. This time corresponds to one period of the deterministic trajectory: the deterministic model is then back to its initial condition. Plot their histograms and evaluate their sample means and sample standard deviations. Identify the deterministic values $x(t^*)$ and $y(t^*)$ on the histograms with `ax.axvline`. Briefly comment on the sample means compared with the deterministic values $x(t^*)$ and $y(t^*)$. (7 points)

d. Repeat part c at $t^* = 15.18$, which corresponds to four periods of the deterministic trajectory, so that the deterministic values are the same as in part c; produce the same histograms. Briefly comment on the sample standard deviations compared with those at $t^* = 3.79$, and on the sample mean of the predators compared with the deterministic value. What does it tell you about the agreement between the deterministic and stochastic predator–prey models over long time horizons? (6 points)

What to hand in

Hand in, per team, in the Brightspace assignment box:

- one Jupyter notebook (`*.ipynb`);
- a PDF export of the executed notebook;
- a Personal Ethics Agreement (see Brightspace);
- an AI Use Declaration (see below).

The notebook is the report. It must contain everything a reader needs to judge your results: the problem set-up, the Python code you ran, the figures, and your conclusions. Presentation matters — use sections, label your axes, and cite any reference outside of the course material.

Put the names of all team members at the top of the notebook, and name the file `A2_` followed by the team members' last names, separated by underscores (e.g. `A2_name1_name2_name3.ipynb`).

Use of artificial intelligence

AI assistance is permitted for this assignment, subject to the documentation requirements below. In contemporary software development, artificial intelligence is increasingly used by programmers to improve productivity, a trend that is likely to continue. Learning to use these tools critically and responsibly is therefore a valuable professional skill. Nevertheless, you must thoroughly understand the concepts and methods on which you are evaluated, particularly because AI tools are not available during in-person evaluations. You remain responsible for verifying, understanding, and being able to explain everything in your submission.

Permitted, properly declared AI use does not reduce your mark. Every submission must contain an **AI Use Declaration**, even if you did not use AI assistance.

1. **Did you use a generative AI tool while completing this assignment?** Answer Yes or No. This includes any type of assistance with code, writing, or editing.

   If you answer **No**, include the following confirmation and omit the remaining questions:

   > *I confirm that I did not use generative AI in preparing this submission.*

   If you answer **Yes**, answer the remaining questions:

2. **How did you use generative AI?** Select all that apply:
   - clarifying concepts or checking your understanding;
   - brainstorming or planning an approach;
   - writing or revising code;
   - debugging code or interpreting error messages;
   - interpreting data, results, tables, or plots;
   - writing or revising explanations;
   - other uses, which you must specify.

3. **Briefly identify the sections or tasks for which AI assistance was used and describe how it influenced your work.**

4. **How did you verify the AI-generated suggestions, code, or explanations?**

5. **Include the following confirmation:**

   > *I confirm that this declaration is complete and that I understand and can explain everything in my submission.*